

Visuo-vestibular Congruency Impacts on Player Experiences in Virtual Reality

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ABSTRACT

Exposure to Virtual Reality (VR) is typically multisensory, incorporating sensory information from several modalities. In this context, the congruency between cross-modal feedback plays a crucial role in enhancing user experience and minimising potential discomfort. Whilst some studies have investigated the visuo-vestibular relationship in VR exposure, the exploration of congruency has often been limited in scope. In this paper, we present early work exploring the role of visuo-vestibular congruency on virtual-reality experiences. This investigation is carried out through a user-based study, evaluating human factors and how they are impacted by congruency conditions. Our initial results highlight some interesting insights, with visually-congruent conditions heightening perceived sickness and presence levels. Furthermore, motion simulation appears to be a useful tool for cybersickness mitigation in older VR users.

Index Terms: Human-centered computing—Virtual reality; Applied computing—Computer games; Computing methodologies—Simulation evaluation; Human-centred computing—User studies

1 INTRODUCTION

Given its application across many domains, understanding the effects of Virtual Reality (VR) on users is a critical topic of research. VR also offers new and granular methods of controlling sensory stimulation, providing a useful tool and platform for multisensory integration research [2]. However, the simulations implemented in VR can bring about sensory experiences which can be strange or uncomfortable to users [8]. For example, cybersickness is a prominent issue in the adoption of virtual reality technology [3]. As such, investigating sensory relationships in VR encompasses a large body of works, with several papers focusing on VR in this context [4]. This is especially true in light of the ever-expanding commercial adoption, and emerging novel technologies such as motion simulation.

Whilst motion simulation has received considerable interest regarding its utility in training, there is comparatively little research in the context of recreation. This is despite the fact that recreational purposes comprise a large number of motion simulation applications. This has been explored in detail [9], with reference to the recent rise in popularity of motion simulators for personal recreation.

The study of sensory relationships during VR activities is another related and important topic of research. For example, cybersickness is a prevalent issue for VR users, and theories suggest this is rooted in conflicting sensory information. For instance, conflicting visuo-vestibular feedback can induce cybersickness [6]. One anecdotal argument for the benefits of motion simulators is that they could potentially mitigate cybersickness by providing vestibular cues matching visual feedback. However, it is largely unknown

how the congruency of visuo-vestibular information could affect recreational experiences. Understanding how types of motion affect a player's experience could lead to insights on how to build game mechanics around motion feedback. Similarly, it could inform a set of guidelines for game developers targeting consumer-level motion simulators. While there are studies that exist which investigate the visuo-vestibular relationship in a motion-actuated setting, most do not consider the congruency of feedback. Those which did found motion cues incongruent to visual feedback considerably induce measures of participant sickness [1].

This early work begins to explore visuo-vestibular congruency in virtual reality in further detail. We investigate congruency in the context of a recreational activity, exploring its impact on player experience and behaviour. This is explored through a user-based study in which participants experience various motion conditions via a motion-actuated driving simulator.

2 STUDY

With these points in mind, a study has been devised using a consumer-available video game and motion simulator. The study was developed to explore the effects of altering correlation in the visuo-vestibular relationship in a VR game. The equipment used throughout the study consisted of the Atomic Motion Systems A3 motion simulator for cued motion and the Oculus Rift VR headset for visual feedback. Additionally, a Thrustmaster TS-XW steering wheel and Fanatec Clubsport shifter peripherals were used for participants to interact with the racing video game. An image of the simulator can be seen in Fig. 1, for further clarity.



Figure 1: A photograph showing the simulator and peripherals used throughout VR exposure.

Three experimental groups have been recruited, each of which differed in the correlation of simulated motion to visual feedback. Visual feedback remained the same, however, the range of motion was altered in each group to affect visuo-vestibular congruency. In one group, motion feedback was positively correlated to visual

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feedback, whereas in the other, the relationship was negatively correlated. The third group was used as a control, which administered no simulated motion throughout. For the sake of brevity, we refer to these as the 'normal motion', 'inverted motion' and 'control' groups, respectively.

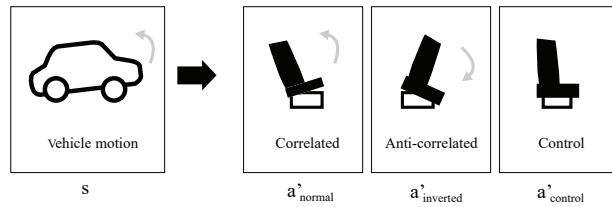


Figure 2: The motion simulator mapping for each of the three motion conditions. The diagram shows a virtual vehicle accelerating.

Each participant will experience all three motion groups, in turn, sampled in a random order without replacement. Random sampling is applied so that each permutation is chosen with equal probability to mitigate the presence of order effects. A within-subjects methodology is opted for to evaluate if treatment group order yielded any effect between participants. In addition, cross-session observations on factors such as skill retention or acclimatisation can be considered in this context. Participants are tasked to complete two laps around a virtual track in a virtual vehicle as quickly as possible. The commercial game 'Project CARS 2' is used as the chosen racing game, as it provides native shared memory support for logging telemetry, and ease of integration with motion simulator software. It is also a popular video game available to the general consumer. An in-game screenshot of Project CARS 2 can be seen in Fig. 3.



Figure 3: An in-game view within Project CARS 2, representative of participant's view throughout VR exposure.

Participants were seated in the motion simulator and briefed on how to interact with the game. The track and virtual vehicle did not change throughout the three sessions and remained the same for all participants. The in-game track/vehicle configuration consisted of the 'Bannochbrae' circuit, with the 'BAC Mono' vehicle. The track was chosen due to its considerable length, with a diverse range of turns for the player to navigate through. This enabled the participant to develop confidence, and provided opportunity to observe effects. Two laps were used to firstly provide an initial lap in which players could re-familiarise themselves, and secondly, to examine how performance was affected between laps.

A total of 48 adult participants were recruited. The majority of participants recruited were between 18-30 years of age, with 77% in the 18-30 group, 8% in the 31-40 group, 10% in the 41-50 group and 4% in the 51-60 group. Furthermore, 33% of participants reported their gender as Female, whilst 66% were Male. Following virtual reality exposure in each session, participants completed the standard

Presence Questionnaire (PQ) [10] and Simulator Sickness Questionnaire (SSQ) [5]. These measures were recorded to gauge the three motion congruency groups impacts on measures of player presence and sickness in post-exposure. Prior to the first session, participants also completed a questionnaire concerning prior experience, demographic information and pre-screening questions to remove those particularly to susceptible to motion sickness. For example, candidates with existing vestibular conditions were screened for participation, as exposure could exacerbate their discomfort.

3 CONCLUSION

This paper presents early work exploring the role of visuo-vestibular congruency on virtual-reality experiences was investigated through a user-based study. The study will evaluate human factors (such as Presence or Sickness) and how they are impacted by different sensory congruency conditions in a VR racing game.

Early results indicate that visually-congruent condition heightened perceived sickness but also improved player presence. Furthermore, it appears that motion also reduced sickness for older participants, potentially highlighting the use of motion simulation to aid in cybersickness mitigation for VR users. Conversely, the visually incongruent motion condition lowered both sickness and presence slightly in comparison to the visually-congruent condition.

Furthermore, there is also a strong relationship between PQ and SSQ measures regardless of motion conditions; participants who are more immersed in the virtual world recorded higher levels of sickness. Similar effects have been observed previously in the literature, with direct correlations between spatial presence and induced vection [7]. Motion actuation increased participant presence but appears to bring about sickness as a byproduct. This could highlight an interesting hurdle for games supporting motion simulation; the inclusion of motion simulation ultimately increases presence, but at a cost to player sickness levels. Investigating measures to mitigate this increase in sickness would be an area of interest for researchers.

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